

RackIQ

Abstract—RackIQ automates the detection, segmentation, text extraction, and processing of medicine strips from images to streamline inventory management and minimize human errors in pharmacies. The system processes an image containing one or more medicine strips through a workflow integrating multiple cutting-edge models and techniques. Initially, YOLO detects and counts the strips, followed by segmentation using the Segment Anything Model (SAM) to isolate individual strips. Text from each strip is then extracted using Microsoft Azure Vision Studio, which accurately captures text without requiring multiple orientations. The extracted text is processed by GPT-4o Mini, generating a structured dictionary detailing the medicine's name, composition, manufacturing and expiry dates, and other essential attributes. Additionally, the system estimates the number of tablets in each strip by calculating the strip's area using a coin-based approach. This structured output supports inventory management and enhances accuracy in identifying medicines and tablet counts, significantly reducing human error. RackIQ's robust pipeline integrates YOLO for object detection, SAM for segmentation, Azure Vision Studio for OCR, and GPT-4o Mini for structured information generation, providing a comprehensive solution for inventory automation.

Index Terms—Segment Anything Model (SAM), YOLO, OCR

I. INTRODUCTION

Efficient and accurate management of pharmaceutical inventories is a critical requirement for healthcare providers, pharmacies, and medical supply chains. Manual methods of tracking medicines, verifying their details, and ensuring inventory accuracy are time-consuming and prone to human errors, which can lead to serious consequences, such as incorrect medication dispensation or stock mismanagement. With the growing complexity of pharmaceutical inventories, there is a pressing need for automated systems that can streamline these processes, reducing the reliance on human labor and improving overall efficiency and accuracy.

RackIQ addresses these challenges by providing a comprehensive, automated solution for detecting, extracting, and analyzing the information contained on medicine strips. This system integrates advanced computer vision techniques, Optical Character Recognition (OCR), and natural language processing to automate the identification of medicines, extract critical information from medicine packaging, and count the number of tablets in each strip. The proposed system leverages several cutting-edge models and algorithms to achieve its objectives. It begins by using the YOLO (You Only Look Once) object detection model, which is well-known for its ability to perform real-time detection with high accuracy. YOLO identifies and

counts the number of medicine strips present in an image, regardless of orientation or background noise.

Following the detection phase, the Segment Anything Model (SAM), which is based on Vision Transformers (ViT), is employed to segment the detected medicine strips from the image. SAM generates precise segmentation masks, allowing the system to isolate each strip for further processing. Segmentation is crucial in separating the strips from irrelevant background elements and preparing them for text extraction.

The text extraction phase utilizes EasyOCR, a widely used OCR tool, to read the text printed on the medicine strips. Medicine strips are often placed at various angles, and their text orientation can differ. To mitigate this, the system rotates the segmented strips in four different orientations, ensuring that the text is accurately captured regardless of the original positioning. The extracted text may include important information such as the medicine name, composition, manufacturing and expiry dates, and other relevant details.

Once the raw text is extracted, it is passed through a custom post-processing module powered by GPT-4o Mini. This natural language processing model processes the OCR-extracted text and generates a structured output in the form of a dictionary. The structured data includes critical attributes such as the medicine name, ingredients, expiry date, and manufacturing date. This organized output significantly reduces the risk of human error during inventory handling and ensures that all relevant details are accurately captured.

The system also includes a tablet-counting mechanism to provide accurate information on the number of tablets in a given strip. Using a coin-based area estimation approach, the system calculates the area of each strip and estimates the number of tablets based on predefined configurations. This method adds a layer of precision to the inventory management process by ensuring that the correct tablet counts are recorded.

RackIQ is particularly useful for automating pharmaceutical inventory management in medical stores and hospitals. By removing the need for manual inspection and text entry, the system enhances the efficiency and accuracy of the entire process. Additionally, by reducing the likelihood of human error, the system can also contribute to improved patient safety by ensuring the correct medications are dispensed and tracked throughout the inventory lifecycle.

This paper presents the detailed design, implementation, and evaluation of the RackIQ. The paper explores

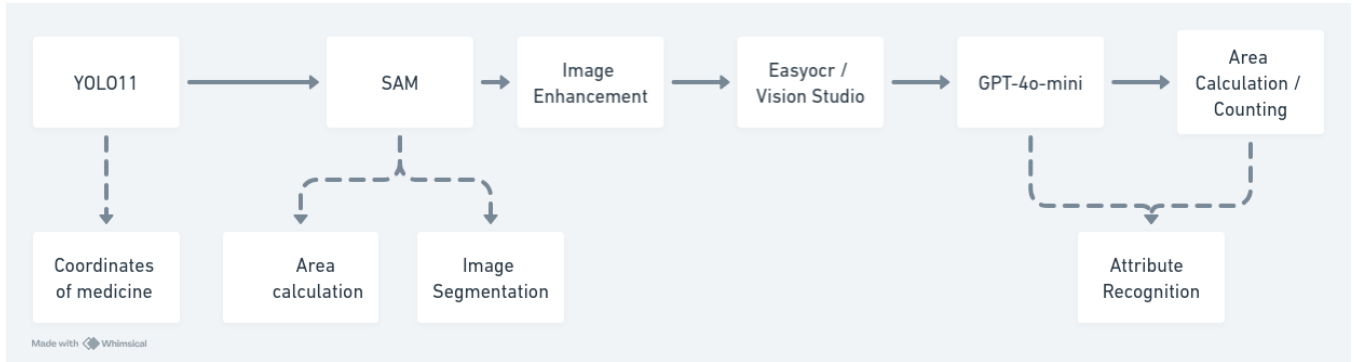


Fig. 1: High Level workflow of RackIQ

the technical components, such as the YOLO object detection model, the SAM segmentation model, OCR-based text extraction, and GPT-based text post-processing, and discusses their roles in ensuring the success of the system. Moreover, this research highlights the importance of integrating deep learning and computer vision techniques in healthcare applications, particularly in automating tedious and error-prone tasks.

II. LITERATURE SURVEY

A. Background

The development of the RackIQ draws upon several cutting-edge advancements in computer vision, deep learning, and natural language processing. In this section, we review the key literature that forms the foundation for the system, specifically focusing on models and techniques for object detection, segmentation, optical character recognition (OCR), and post-processing of extracted data using language models.

B. Literature Survey

The YOLO (You Only Look Once) family of models has revolutionized object detection by achieving real-time performance while maintaining high accuracy. YOLO models are designed to detect multiple objects in a single pass through a neural network, which contrasts with region-based approaches like R-CNN, where multiple passes are required for object proposals. Redmon et al. introduced the YOLO model in their 2016 paper, “You Only Look Once: Unified, Real-Time Object Detection” [Redmon et al., 2016], where they emphasized a single network architecture that directly predicts bounding boxes and class probabilities from the input image.

The primary advantage of YOLO is its speed, as it divides the image into a grid and applies convolutional neural networks to the entire image rather than focusing on specific regions of interest. This makes YOLO particularly suitable for applications that require real-time performance, such as detecting multiple medicine strips in the proposed system. Successive iterations of YOLO, such as YOLOv3, YOLOv4, and YOLOv5, have improved upon the accuracy and efficiency of the original model by

incorporating techniques such as better feature pyramids, deeper architectures, and more sophisticated anchor box mechanisms. In this project, YOLOv8 was employed for object detection due to its high mean Average Precision (mAP) values across various applications.

Segmentation plays a crucial role in the RackIQ, as it allows for the precise isolation of medicine strips from complex backgrounds. The Segment Anything Model (SAM), introduced by Kirillov et al. in their 2023 paper, “Segment Anything,” has emerged as a powerful and versatile tool for image segmentation. SAM is built on the Vision Transformer (ViT) architecture, which enables it to generate segmentation masks by taking into account both local and global information in the image.

SAM’s key strength lies in its ability to perform segmentation on any object in an image without requiring task-specific annotations or training data. This makes it ideal for scenarios where manual annotation is impractical, such as handling large-scale pharmaceutical inventories. By leveraging SAM, the RackIQ ensures that only relevant objects (i.e., medicine strips) are segmented, while the background and other irrelevant elements are ignored. This model’s generalizability and scalability are valuable in medical settings, where variability in packaging, lighting, and strip arrangement can pose challenges for traditional segmentation methods.

Accurate text extraction from medicine strips is essential for identifying key attributes such as the medicine name, composition, and expiry date. OCR technology has undergone significant advancements, enabling it to recognize printed and handwritten text in images with higher precision. EasyOCR, one of the prominent OCR tools, is utilized in this project for text extraction due to its capability to handle a wide variety of fonts, languages, and text orientations.

The OCR field has its roots in classical image processing techniques, but deep learning-based approaches, particularly convolutional neural networks (CNNs), have significantly improved the accuracy and robustness of text recognition. Long et al. in their 2018 paper “Scene Text Detection and Recognition: The Deep Learning Era” [Long et al., 2018], provided a comprehensive review of

how CNN-based architectures have replaced traditional methods in OCR tasks, especially in real-world scenarios like street sign recognition and document digitization.

In the context of this project, the medicine strips can appear at different angles, and text orientation can vary significantly. Therefore, the system performs multiple rotations of the segmented strips to ensure accurate text extraction from all orientations. OCR accuracy is further enhanced through pre- and post-processing steps that correct common misclassifications and ambiguities in the recognized text.

III. METHODOLOGY

A. Data Collection

For this project, we collected a dataset of 1200 images comprising approximately 80 different types of medical strips. The images were captured using a mobile camera with a resolution of 1980x2000 pixels. The data was obtained under controlled conditions on a white platform, with a distinctive black circle positioned at the top right corner of the frame. This circle will serve as a reference for estimating the area of the strips in later stages of the analysis, particularly for tablet counting.

The dataset includes both single-strip images and images containing multiple strips, either of the same or different types. By ensuring variability in the number and arrangement of strips, the dataset is designed to simulate real-world conditions in medical inventory scenarios.

1) *Architecture of YOLOv8*: YOLOv8 (You Only Look Once version 8) is a state-of-the-art object detection model that builds upon the strengths of previous YOLO models. It operates by dividing an image into grids and predicting bounding boxes, objectness scores, and class probabilities in a single pass. YOLOv8 utilizes a convolutional neural network (CNN) backbone, feature pyramid network (FPN), and anchor-based detection heads for efficient object detection across multiple scales. The architecture is designed to achieve a balance between detection accuracy and speed, making it well-suited for real-time object detection tasks.

2) *Training and Fine-Tuning Process*: Initially, we selected YOLOv8 due to its strong performance in detecting objects in various settings, including single and multiple medical strips on different backgrounds. We fine-tuned YOLOv8 on our custom dataset of 800 images of medical strips using transfer learning techniques. The model was trained on a pre-existing YOLOv8 checkpoint and fine-tuned on our dataset using the Ultralytics library.

The model performed well on most images, but during evaluation, we observed that YOLOv8 had difficulties in accurately detecting tilted or rotated medical strips. This led to errors in detecting strips that were at non-standard angles or with slight obstructions. Therefore, to address these challenges, we switched to YOLO11.

3) *Transition to YOLO11*: YOLO11, an advanced version of YOLO, provided enhanced detection performance for medical strips in challenging conditions, particularly for

tilted strips and strips positioned at odd angles. YOLO11 incorporates improvements such as adaptive feature pooling and a more robust backbone, which significantly improves its ability to detect objects under a variety of orientations. The key motivation for using YOLO11 over other detection technologies was its exceptional ability to handle angle variations in object detection, which was crucial for our dataset where strips are often rotated.

We fine-tuned YOLO11 using pre-trained weights (yolo11n.pt) from the Ultralytics repository. The model was trained for several epochs, and the performance was compared to YOLOv8 across multiple evaluation metrics.

4) *Evaluation Metrics*: To assess the performance of both YOLOv8 and YOLO11, we used the following evaluation metrics:

- **Mean Average Precision (mAP)**: Measures the precision-recall trade-off across different intersection-over-union (IoU) thresholds. We report both mAP50 (IoU threshold of 50%) and mAP50-95 (average IoU thresholds from 50% to 95%).
- **Accuracy**: The proportion of correctly detected medical strips.
- **Precision and Recall**: Precision indicates how many of the detected strips were correct, while recall measures how many true strips were detected.

The results for both models are summarized in Table I:

TABLE I: Performance Metrics for YOLOv8 and YOLO11

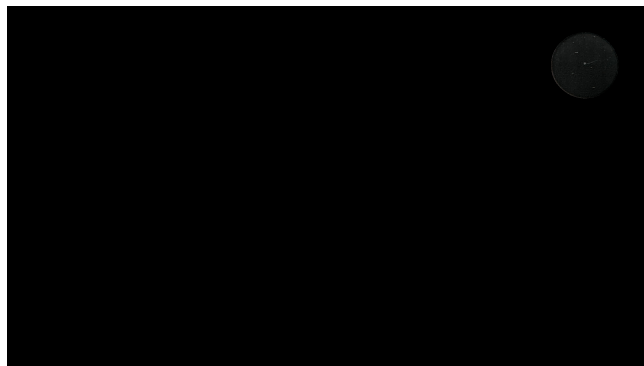
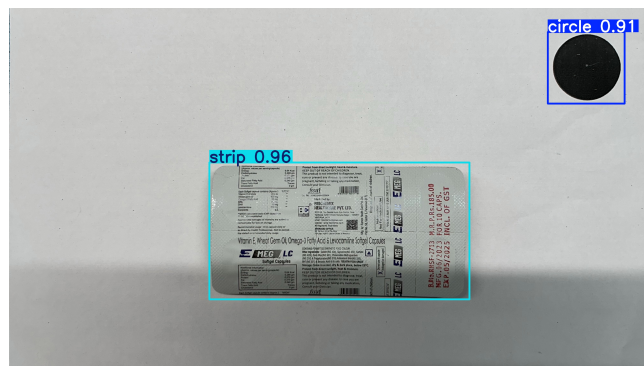
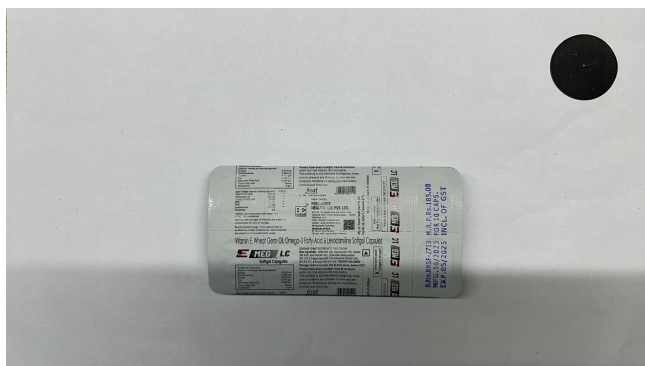
Metric	YOLOv8	YOLO11
Accuracy	0.98	0.998
mAP50	0.92	0.99
mAP50-95	0.86	0.95

YOLO11 consistently outperformed YOLOv8 in terms of accuracy and mean average precision, particularly for images with tilted or rotated strips. This improvement made YOLO11 the optimal choice for our object detection task, as it addressed the critical issue of accurately detecting strips at various orientations, which was a key requirement for further segmentation and text extraction steps.

B. Strip Segmentation using SAM (Segment Anything Model)

1) *Integration of SAM for Medical Strip Segmentation*: The Segment Anything Model (SAM) is a powerful segmentation model capable of segmenting any object in an image based on provided prompts, such as bounding boxes. In our system, SAM is integrated for the precise segmentation of medical strips detected by YOLO11. Once YOLO11 detects the medical strips and the reference circle used for area estimation, it provides the bounding box coordinates of each strip and circle in the image. These coordinates are then passed to SAM, which segments the objects within those bounding boxes.

This integration between YOLO11 and SAM provides a significant advantage. YOLO11 detects the general location

[illegible]

```
{
  'Medicine_Name': 'Meg',
  'Formula': 'Vitamin E, Omega-3 Fatty Acid, Levocarnitine',
  'Manufacturer': 'Healthcare Pvt. Ltd.',
  'Dosage': '1 capsule daily',
  'Strength': '1 capsule',
  'Expiry_date': '05/2025',
  'Manufacture_date': '06/2023',
  'Price': '185.00'}
```

Fig. 2: Output Images at Diifferent Stages of Workflow

of the medical strips and circles, and SAM refines the segmentation by focusing only on the regions within the bounding boxes. This approach enhances the speed and efficiency of the segmentation process since SAM only processes the relevant areas rather than the entire image.

2) *Post-Processing of Segmentation Results:* After receiving the bounding box coordinates from YOLO11, SAM returns three segmented components: the medical strip, the bounding box enclosing the strip, and the reference circle. These segmented objects are further processed for downstream tasks, such as text extraction and area-based counting.

- **Segmented Strip:** SAM accurately segments the medical strip inside each bounding box. This segmented strip is crucial for subsequent stages, such as OCR text extraction, which is used to identify the medicine details from the strip. The bounding box coordinates are refined by SAM to improve the precision of object

boundaries. This ensures that the strip segmentation is tightly coupled with the object, reducing the chances of extracting irrelevant parts of the image.

- **Segmented Circle:** SAM also segments the black reference circle, which is used later for area estimation and tablet counting based on size comparison with the circle.

3) Efficiency Gains from YOLO and SAM Integration:

The use of YOLO11 for object detection followed by SAM for precise segmentation provides an efficient pipeline. YOLO11 quickly identifies the potential regions of interest, and SAM performs a fine-grained segmentation within those areas. This combination makes the system faster and more accurate than using SAM alone for the entire image. The bounding box coordinates from YOLO11 allow SAM to focus its efforts on specific objects, minimizing computational overhead while ensuring high-quality segmentation results.

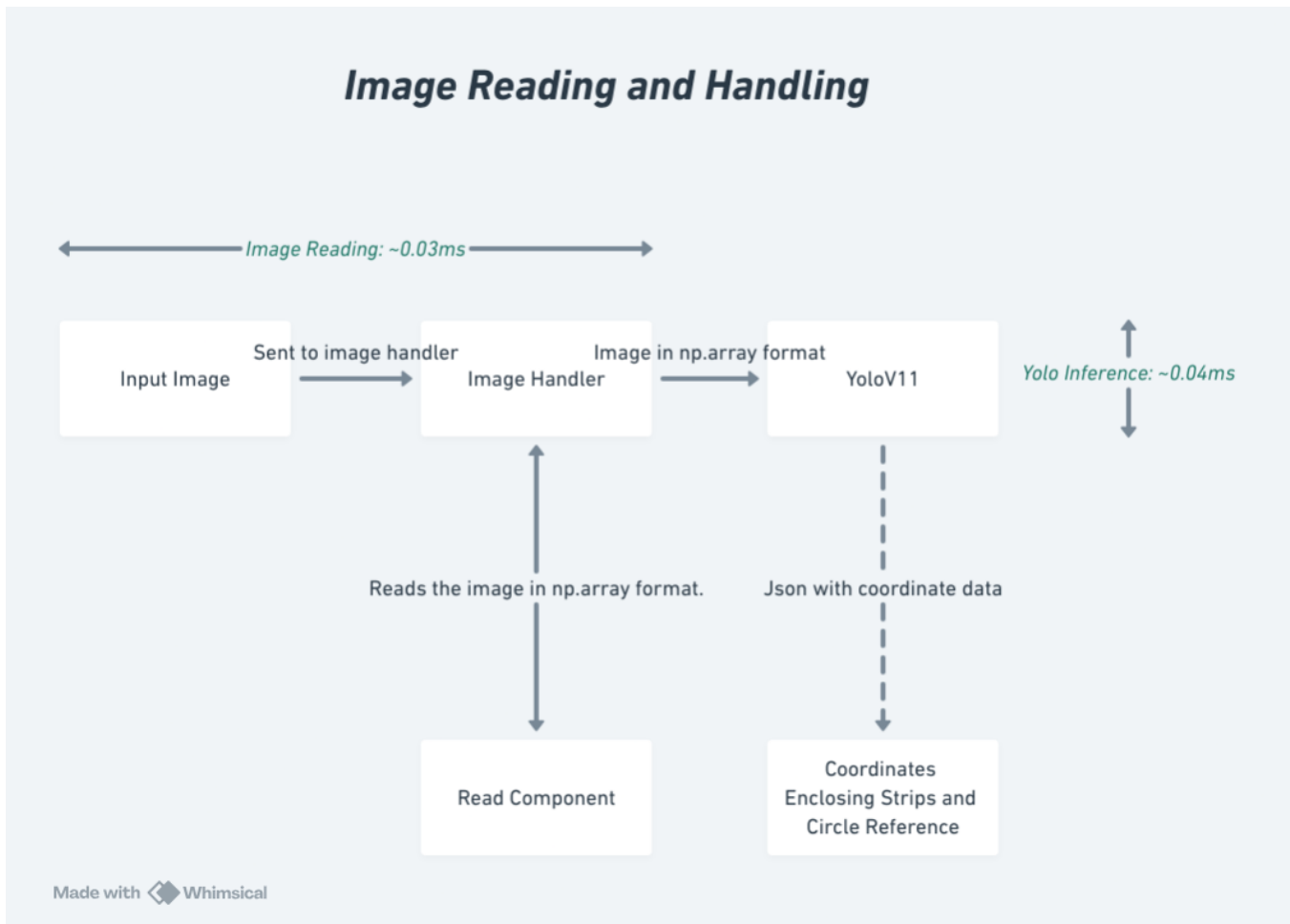


Fig. 3: Workflow of YOLO11

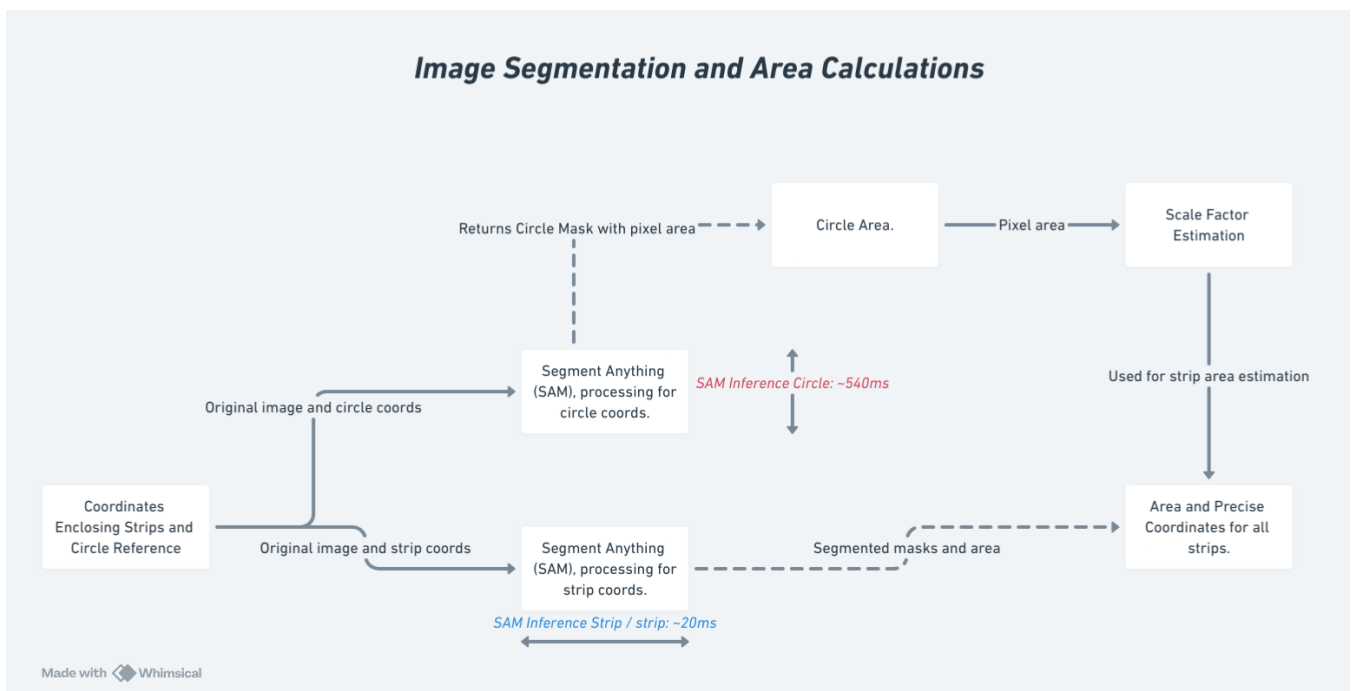


Fig. 4: Workflow of SAM

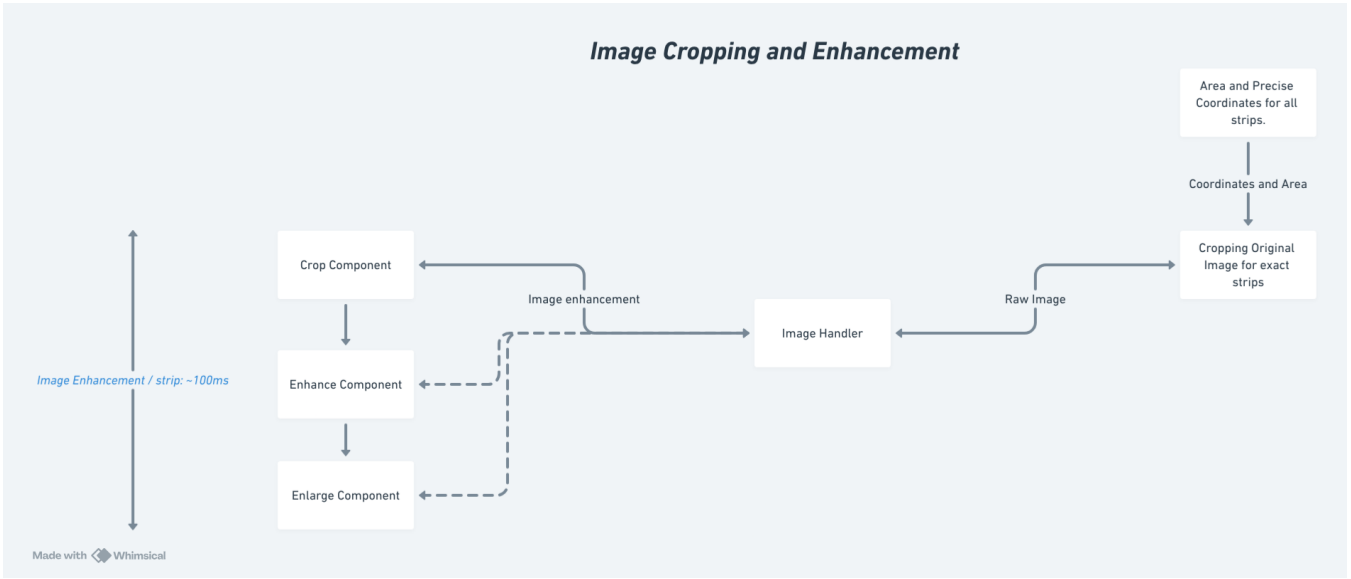


Fig. 5: Image enhancement

By leveraging the capabilities of both YOLO11 and SAM, we achieve a robust and efficient segmentation process, which is critical for the accurate detection and analysis of medical strips in real-world conditions.

C. Optical Character Recognition (OCR)

1) *Techniques for Text Extraction:* For extracting text from segmented medical strips, we initially used the open-source EasyOCR library. EasyOCR relies on the CRAFT (Character Region Awareness for Text Detection) model for text detection and employs the deep-text-recognition-benchmark for text recognition, using the 'None-VGG-BiLSTM-CTC' architecture. This fully convolutional network allowed for variable text length predictions, making it effective for aligned text. However, challenges arose when text on medical strips was rotated or tilted, impacting extraction accuracy.

To address these limitations, we transitioned to Microsoft Azure Vision Studio for OCR, which demonstrated superior accuracy and consistency. Microsoft Azure Vision Studio effectively handled text extraction across various orientations without requiring additional preprocessing steps for rotation. This improved the overall text sanity, ensuring reliable extraction even when text was printed at unconventional angles on the strips.

2) *Handling Multiple Text Orientations:* Medical strips often contain text printed in various orientations, which initially posed challenges for text extraction.

With EasyOCR, we had implemented a multi-angle extraction approach, rotating each strip at 90°, 180°, and 270° angles to capture text from different orientations, then combining the results to achieve accurate extraction. However, after adopting Microsoft Azure Vision Studio, this rotation step was no longer necessary. Azure Vision Studio's advanced OCR capabilities efficiently handled multi-orientation text directly, streamlining the process

and significantly improving the accuracy and completeness of the extracted information.

3) *Post-Processing with GPT-4o Mini:* Despite the rotation-based approach, the raw text extracted by EasyOCR or Vision Studio was still often clumsy and contained inaccuracies. To rectify these errors and obtain meaningful text, we employed the GPT-4o Mini model for post-processing.

The combined text from EasyOCR's multiple orientations or vision studio was provided as input to GPT-4o Mini, which processed the text and generated a structured output. This post-processing step involved correcting OCR inaccuracies, resolving issues with incomplete words, and structuring the data into meaningful information such as medicine names, expiration dates, and other important attributes.

By leveraging GPT-4o Mini for post-processing, we significantly enhanced the quality and usefulness of the extracted text, transforming it from a raw, error-prone output into a reliable, structured format for downstream analysis.

D. Text Processing and Attribute Inference using GPT-4o Mini

The GPT-4o Mini model was employed for structured attribute inference from the OCR output of the medicine strips. After obtaining the text from the EasyOCR or vision studio output, the model was prompted to extract relevant attributes such as the medicine name, composition, expiry date, manufacturing date, and other key details. This process is crucial for accurately updating the medical inventory based on the extracted information.

1) *Prompt Structure:* To ensure accurate extraction of attributes, the GPT-4o Mini model was prompted with a carefully crafted instruction that framed the context.

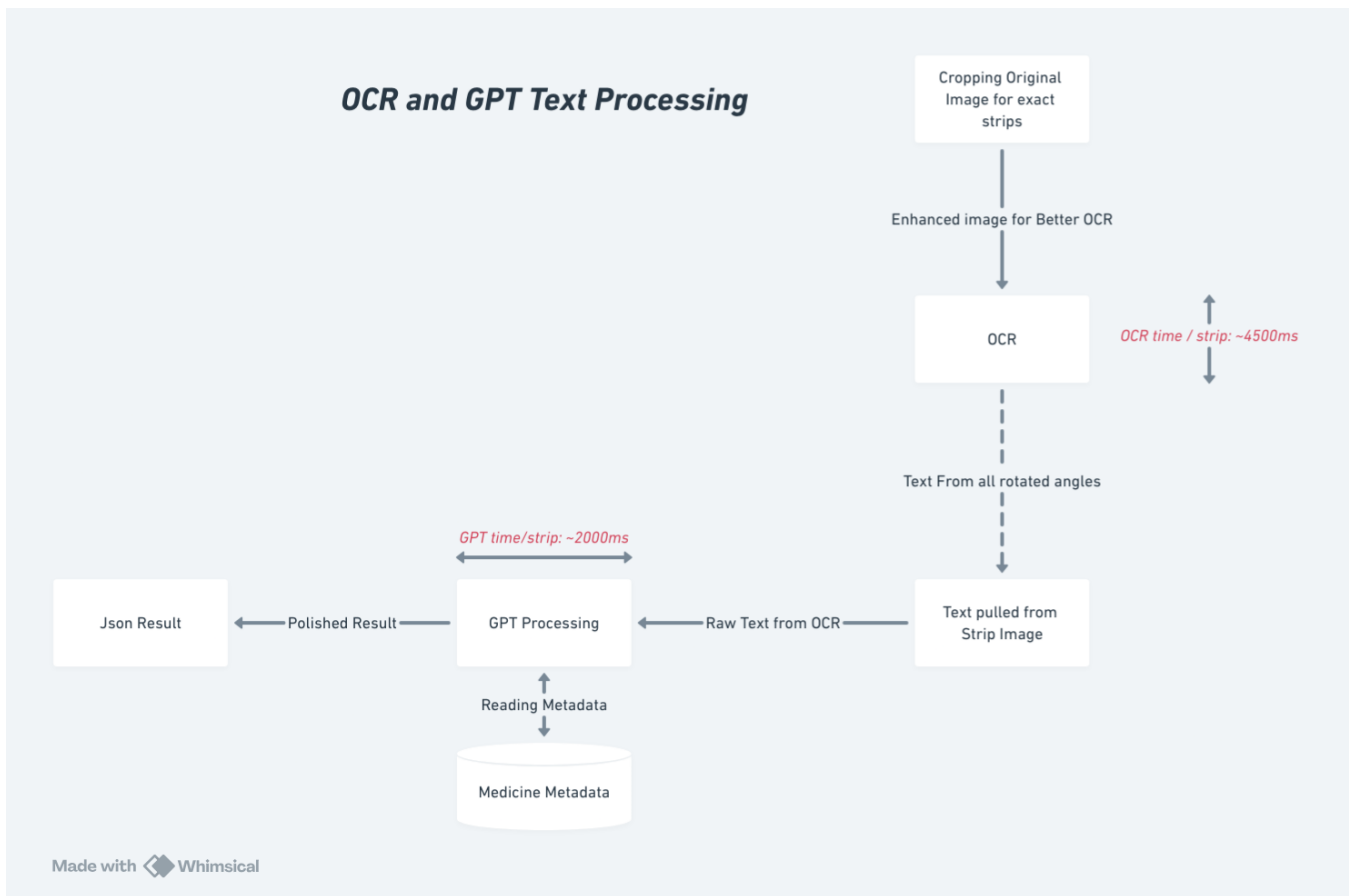


Fig. 6: Text Extraction and Post-processing with GPT

The following prompt was used to guide the model in generating the structured output:

"You are a medical expert and also a JSON expert. You will be given a text which is the OCR output from the medicine strip image. These are the possible medicine names: {medicine_strip_names} that you can expect in the OCR output. You are expected to extract the following attributes: Medicine Name, Composition, Expiry Date, Manufacturing Date, and any other relevant details. Please return the result as a JSON object."

This prompt ensures that the model understands both the medical context and the required output format (JSON). The list of possible medicine names is passed dynamically based on the strips detected in the image.

2) *Key Attributes Extracted:* The following attributes were extracted and stored in a structured JSON format:

- **Medicine Name:** The name of the medicine found in the OCR output.
- **Composition:** Key ingredients or chemical composition of the medicine.
- **Expiry Date:** The expiry date printed on the strip.
- **Manufacturing Date:** The manufacturing date printed on the strip.
- **Other Details:** Any additional information such as

dosage or storage instructions.

3) *Example Output:* Given the OCR text as input, the following is an example of the JSON output generated by GPT-4o Mini:

```
{
  "Medicine Name": "Paracetamol",
  "Composition": "Paracetamol 500mg",
  "Expiry Date": "2025-06",
  "Manufacturing Date": "2023-03",
  "Other Details": "Store in a cool, dry place"
}
```

This structured data can be directly used for updating the inventory system, enabling efficient management of medical strips.

4) *Handling Multiple Orientations:* Initially, to improve OCR accuracy, text extraction was performed after aligning the strips in multiple orientations. With EasyOCR, output from all four rotations was passed through the GPT-4o Mini model, and the best match for attributes was selected based on confidence scores.

However, after transitioning to Microsoft Azure Vision Studio, the need for multiple rotations was eliminated. Azure Vision Studio's robust OCR capabilities accurately extracted text from various orientations without addi-

tional alignment, enhancing both efficiency and accuracy. The output was then processed through the GPT-4o Mini model to identify attributes, now benefiting from increased confidence in text extraction accuracy across diverse orientations.

5) *Challenges and Methods for Attribute Recognition:* One of the primary challenges during the attribute recognition process was the variability in the OCR output quality. Due to text misalignment, blurred images, or varying lighting conditions, the OCR text could sometimes be incomplete or inaccurate. This posed difficulties in correctly identifying key attributes such as the medicine name or expiry date.

To mitigate this, we ensured that the input text provided to GPT-4o Mini was pre-processed by extracting text from multiple orientations and combining it, as explained in the previous section. Additionally, the model was fine-tuned to focus on the specific format of medical strips, which helped it better recognize patterns like the expiry date format or the common structure of medicine formulas.

GPT-4o Mini's ability to infer missing or unclear information, such as partially recognized medicine names or ambiguous expiry dates, further enhanced its effectiveness in attribute recognition. The result was a robust system capable of generating a comprehensive dictionary of attributes for each medical strip, despite the challenges posed by real-world data variability.

E. Tablet Counting using Coin-based Area Estimation

In our approach to count the number of tablets in a medical strip, we employed a coin-based area estimation method. A known reference object, a circle (coin) with a fixed real-world area, was included in each image to provide a reliable scale factor. The process of estimating the tablet count consists of the following steps:

- **Circle Area Calculation:** First, we segmented the circle in the image using the SAM (Segment Anything Model). The area of the circle in pixels was calculated by summing up the pixels (ones) inside the segmented region.
- **Scale Factor Calculation:** Using the known real-world area of the coin and the pixel area calculated, we derived the scale factor, which is the ratio of the real-world area to the pixel area.
- **Strip Area Calculation:** Similarly, the medical strip was segmented using SAM, and its area in pixels was computed by summing the pixel (ones) values within the segmented region.
- **Tablet Counting:** Using the calculated scale factor, we converted the pixel area of the strip to its real-world area. Knowing the approximate area of one tablet, we then divided the total area of the strip by the area of a single tablet to estimate the total number of tablets in the strip.

It is important to note that the area of the circle and the strip were calculated once in the segmented parts

provided by SAM. This method provided an accurate and efficient way to estimate the number of tablets in various orientations and sizes of medical strips.

1) *Methodology for Using Reference Coin for Area Estimation:* To estimate the number of tablets in a medical strip, we employed a coin-based area estimation technique utilizing a reference coin with a known actual area of 7.06cm^2 . The following steps outline the methodology:

- 1) **Calculate the area of the coin in pixels:**

$$A_{\text{coin}}^{\text{pixels}} = \text{Area of the coin in pixels (segmented circle)}$$

- 2) **Calculate the scaling factor:**

$$S = \frac{A_{\text{coin}}^{\text{actual}}}{A_{\text{coin}}^{\text{pixels}}}$$

$$\text{where } A_{\text{coin}}^{\text{actual}} = 7.06\text{cm}^2.$$

- 3) **Calculate the area of the strip in pixels:**

$$A_{\text{strip}}^{\text{pixels}} = \text{Area of the strip in pixels (segmented image)}$$

- 4) **Convert the pixel area of the strip to real-world area:**

$$A_{\text{strip}}^{\text{cm}^2} = A_{\text{strip}}^{\text{pixels}} \times S$$

- 5) **Retrieve the actual area of the full strip from metadata:**

$$A_{\text{strip}}^{\text{actual}} = \text{Area of full strip from metadata}$$

- 6) **Determine the portion of the strip in the image:**

$$P_{\text{strip}} = \frac{A_{\text{strip}}^{\text{cm}^2}}{A_{\text{strip}}^{\text{actual}}}$$

- 7) **Estimate the number of tablets:**

$$N_{\text{tablets}} = P_{\text{strip}} \times T_{\text{full strip}}$$

$$\text{where } T_{\text{full strip}} = \text{Number of tablets in the full strip.}$$

2) *Algorithm for Tablet Counting:* The algorithm for counting tablets using the coin-based area estimation is as follows:

Algorithm 1: Tablet Counting Algorithm

```

1: Input: Image of the strip, known area of the coin,
   metadata of the strip Output: Estimated number of
   tablets CountTabletsImage,  $A_{\text{coin}}^{\text{actual}}$ , Metadata
 $A_{\text{coin}}^{\text{pixels}} \leftarrow$  Calculate area of the coin in pixels
 $S \leftarrow \frac{A_{\text{coin}}^{\text{actual}}}{A_{\text{coin}}^{\text{pixels}}}$ 
 $A_{\text{strip}}^{\text{pixels}} \leftarrow$  Calculate area of the strip in pixels
 $A_{\text{strip}}^{\text{cm}^2} \leftarrow A_{\text{strip}}^{\text{pixels}} \times S$ 
 $A_{\text{strip}}^{\text{actual}} \leftarrow$  Retrieve area of full strip from metadata
 $P_{\text{strip}} \leftarrow \frac{A_{\text{strip}}^{\text{cm}^2}}{A_{\text{strip}}^{\text{actual}}} \quad T_{\text{full strip}} \leftarrow$ 
Get number of tablets in full strip from metadata
 $N_{\text{tablets}} \leftarrow P_{\text{strip}} \times T_{\text{full strip}}$  return  $N_{\text{tablets}}$ 

```

IV. SYSTEM ARCHITECTURE

A. Overview of the System Pipeline

The proposed system leverages a modular architecture to streamline the process of inventory management for medical strips. The pipeline begins with the capture of images of medical strips, which are then processed through various modules, including object detection, segmentation, optical character recognition (OCR), and text processing.

B. Component-level Breakdown

The system consists of several critical components, each contributing to the overall functionality of the application. Below is a detailed description of each module, along with the software and hardware specifications used for implementation.

1) *YOLO (You Only Look Once)*: YOLO is employed for real-time object detection of medical strips in the captured images. The model is trained to identify strips accurately, providing bounding box coordinates that are used in the subsequent segmentation process.

2) *SAM (Segment Anything Model)*: The SAM is integrated for effective segmentation of the detected medical strips. By utilizing the bounding box coordinates provided by YOLO, SAM accurately segments the strips from the background. This model significantly enhances the visibility of the strips for text extraction.

3) *Microsoft Azure Vision Studio*: Microsoft Azure Vision Studio is used for extracting text from the segmented strips. It utilizes advanced OCR techniques to accurately recognize text across diverse fonts and layouts. This model has proven robust for handling multiple text orientations, eliminating the need to rotate images at different angles. Azure Vision Studio ensures comprehensive and accurate text capture directly, improving the reliability of extracted data.

4) *GPT-4o Mini*: The extracted text from Microsoft Azure Vision Studio is then processed using GPT-4o Mini, which structures the raw OCR output into a well-defined dictionary containing key attributes such as medicine name, formula, strength, expiry date, and manufacturer details. This integration enables seamless conversion of raw text into organized data for further processing.

5) *Hardware and Software Specifications*: The implementation is executed on an AWS server utilizing NVIDIA A10 instances, specifically configured as follows:

- **vCPUs**: 16
- **Memory**: 32 GiB
- **Memory per vCPU**: 2 GiB
- **Physical Processor**: AWS Graviton2 Processor
- **Clock Speed**: 2.5 GHz
- **CPU Architecture**: arm64
- **GPU**: 1
- **GPU Architecture**: NVIDIA T4G Tensor Core
- **Video Memory**: 16 GiB
- **GPU Compute Capability**: 7.5

The total RAM consumption during operation is summarized below:

- **YOLO** ~ 400 MB
- **EasyOCR** ~ 300 MB
- **SAM (loading on GPU)** ~ 4.0 to 4.5 GB
- **SAM (inference)** ~ 1.5 GB

The overall total RAM utilized by the system ranges from 7 to 7.5 GB, ensuring efficient processing of images and data without exceeding the available resources.

6) *Python Libraries Used*: The implementation of the system utilizes the following Python libraries:

- `easyocr==1.7.1`
- `Flask==3.0.3`
- `numpy==2.1.2`
- `openai==1.51.2`
- `opencv_python==4.10.0.84`
- `opencv_python_headless==4.10.0.84`
- `pandas==2.2.3`
- `Pillow==10.4.0`
- `python-dotenv==1.0.1`
- `segment_anything==1.0`
- `streamlit==1.38.0`
- `torch==2.2.0`
- `ultralytics==8.3.3`

V. EXPERIMENTS AND RESULTS

A. Object Detection Performance

The performance of the YOLO11 model was evaluated using the Mean Average Precision (mAP) metric, which was found to be 0.94. This metric highlights the model's ability to accurately detect medical strips in various conditions. The confidence score for the Segment Anything Model (SAM) ranged between 0.98 and 0.995, indicating a high level of precision in the segmentation process.

We conducted experiments on different image types, considering factors such as lighting conditions and strip orientations. The following table summarizes the inference times for each step in the pipeline for selected images:

TABLE II: Inference Time Ranges for Each Component

Component	Time Range (s)
Image Reading	0.03 - 0.04
YOLO	0.04 - 1.02
Circle Mask Generation	0.54 - 0.73
Strip Mask Generation	0.02 - 0.02
OCR	3.5 - 8.5
GPT	1.38 - 2.08
Total Inference Time	7.00 - 14.00

B. Segmentation Performance

The accuracy of the SAM in segmenting medical strips was evaluated both qualitatively and quantitatively. Qualitative results indicated that the model effectively distinguished the strips from the background under varying lighting conditions. The quantitative results corroborated these findings, demonstrating high segmentation accuracy with an average confidence score of 0.98 to 0.995.

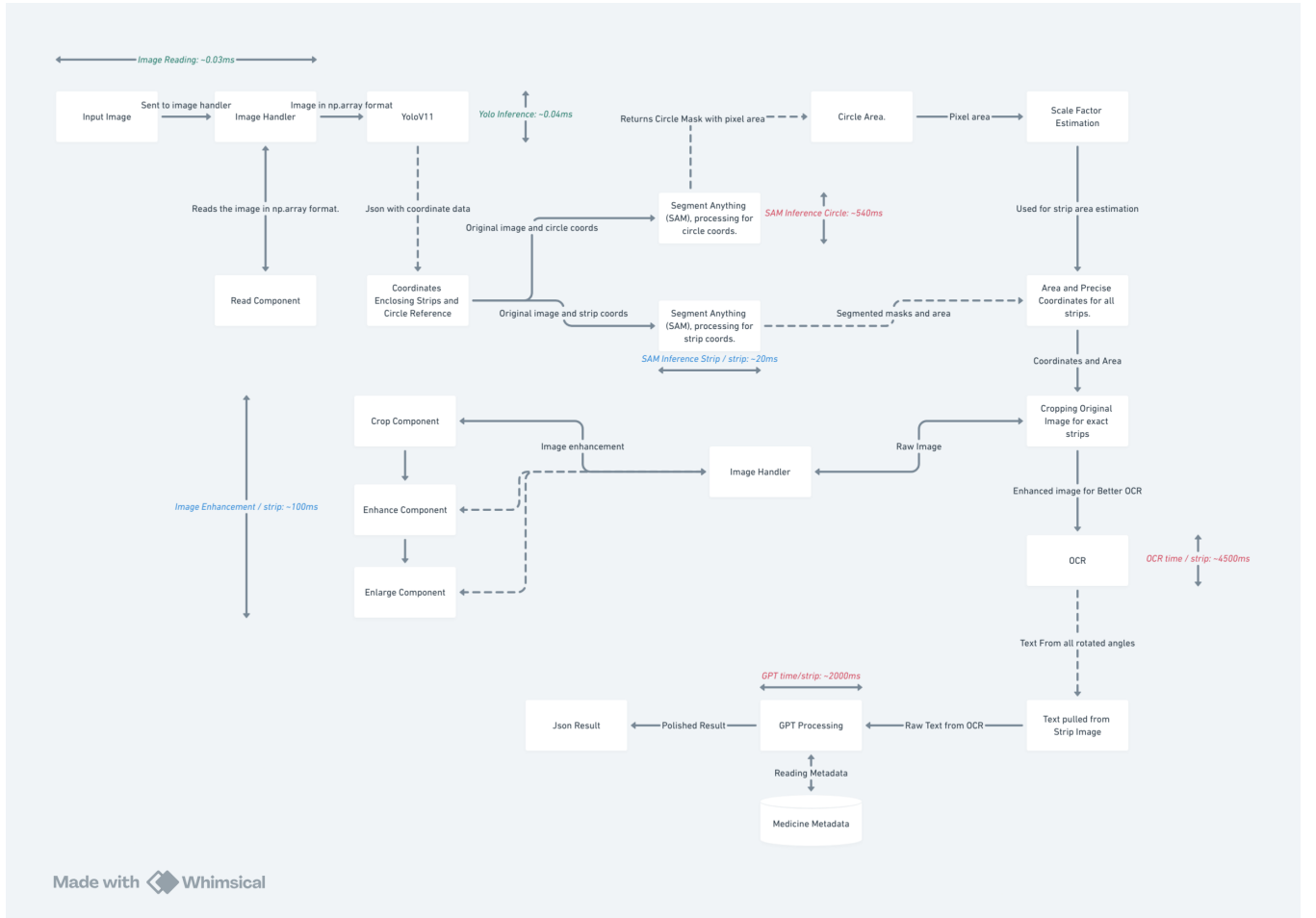


Fig. 7: Workflow of RackIQ

C. OCR Accuracy

The performance of EasyOCR was initially assessed, revealing challenges such as misread characters and difficulties with specific fonts and text orientations. To address these limitations, we implemented a rotation-based approach, extracting text from multiple orientations and aligning strips horizontally to improve accuracy.

However, after transitioning to Microsoft Azure Vision Studio, OCR accuracy improved significantly without the need for manual rotation or alignment. Azure Vision Studio's advanced OCR capabilities effectively handled various text orientations and fonts directly, eliminating the previous challenges and enhancing the reliability and consistency of text extraction.

D. Attribute Recognition

The accuracy of GPT-4o Mini in generating structured outputs from the OCR data was also evaluated. The model effectively translated the raw text into a well-defined dictionary containing essential attributes of the medicine. However, error analysis revealed some inaccuracies in attribute recognition, prompting further refinement of the text processing pipeline to enhance performance.

TABLE III: Attribute recognition Comparison between EasyOCR and Microsoft Azure Vision Studio OCR techniques

Attribute	EasyOCR	Vision Studio
Medicine Name	80.43%	100%
Formula	70%	80%
Manufacturer	61.10%	90%
Dosage	52.10%	80%
Strength	60%	80%
Manufacturing Date	50.40%	100%
Expiry Date	50.20%	100%
Price	52.10%	80%

E. Tablet Counting Accuracy

The precision of the coin-based area estimation method was analyzed. The resolution of the captured images played a crucial role in counting accuracy. By employing a statistical method based on the area estimation of the reference circle, we minimized counting errors, ensuring reliable tracking of the number of tablets.

F Summary of Results

The results indicate a successful integration of YOLO, SAM, EasyOCR, vision studio and GPT-4o Mini in achieving high accuracy in inventory management tasks, in-

cluding object detection, segmentation, OCR, attribute recognition, and tablet counting. Continuous improvement of each component's performance is essential to further enhance the overall system's effectiveness.

VI. CONCLUSIONS

In this paper, we presented an innovative RackIQ that integrates state-of-the-art technologies to enhance the accuracy and efficiency of medical strip analysis. Our approach leverages advanced components, including YOLO for object detection, the Segment Anything Model (SAM) for precise segmentation, Microsoft Azure Vision Studio for optical character recognition, and GPT-4o Mini for generating structured outputs.

The experimental results demonstrate the effectiveness of our system, achieving a mean Average Precision (mAP) of 0.94 for YOLO and a high confidence score of 0.98 to 0.995 for SAM. Transitioning to Microsoft Azure Vision Studio significantly improved OCR accuracy by eliminating the need for rotation-based techniques, as the model handled text extraction across varied orientations directly and reliably. Additionally, the coin-based area estimation method provided a robust solution for accurately counting tablets, minimizing human error and addressing potential discrepancies in inventory management.

This research not only showcases the potential of combining deep learning and natural language processing for practical applications in healthcare but also emphasizes the importance of systematic data processing pipelines in real-world scenarios. The findings of this study contribute to the ongoing development of automated medical systems, paving the way for further innovations in the field.

VII. FUTURE WORK

The RackIQ presents a robust framework for automated medical strip analysis; however, several avenues for future work can further enhance its capabilities and applicability. Key areas of focus include:

- 1) **Extension to Strip Boxes:** We plan to extend the functionality of our system to handle strip boxes

containing multiple strips. This will involve modifying the segmentation and counting algorithms to accurately identify and analyze each strip within the box, improving the system's versatility in different packaging formats.

- 2) **Improved Strip Alignment Algorithms:** Developing a more sophisticated algorithm for strip alignment prior to presenting them to EasyOCR is essential. By enhancing the pre-processing step, we aim to ensure better text extraction quality, particularly for strips that may not be perfectly oriented in the captured images.
- 3) **Metadata Development:** To increase the system's efficacy, we will work on developing comprehensive metadata for a larger number of strips. This will include integrating additional attributes such as dosage information, manufacturer details, and safety warnings, thereby enriching the output provided to users and enhancing the system's utility in clinical settings.
- 4) **Production and Hosting:** Transitioning the system to a production environment is a crucial next step. This involves hosting the application on a cloud platform, conducting extensive testing to identify and address real-time operational issues, and ensuring system reliability and scalability.
- 5) **Handling Cut or Half Strips:** Enhancing the system's capability to accurately analyze half or cut medical strips is another significant focus area. This will require the development of advanced segmentation techniques to ensure that relevant information is captured even when the strips are not intact, thereby broadening the range of scenarios in which the system can be effectively utilized.

By addressing these areas, we aim to develop a more robust, flexible, and user-friendly Intelligent Medical Strip Analysis system that can significantly contribute to improving medication management and reducing human error in medical settings.